67. Let the numbers be x, y and z.

Then, $\left(\frac{x+y}{2}\right)-\left(\frac{y+z}{2}\right)=15$ or $(x + y) - (y + z) = 30$

or $x - z = 30$.

68. Sum of 50 numbers = 38 × 50 = 1900.

Sum of remaining 48 numbers = 1900 – (45 + 55) = 1800.

$\therefore$ Required average = $\left(\frac{1800}{48}\right)=37.5$.

69. Excluded number = (18 × 5) – (16 × 4) = 90 – 64 = 26.

70. New observation = (47 × 7) – (45.5 × 6) = 329 – 273 = 56.

71. Sum of the ages of 24 students = (15 × 30) – (16 × 6) = 450 – 96 = 354.

$\therefore$ Required average = $\left(\frac{354}{24}\right)=14\frac{3}{4}$ yrs = 14 yrs 9 mths.

72. Required average = $\left[\frac{(40\times 7)-(39\times 4)}{3}\right]^{\circ}$C

$=\left(\frac{124}{3}\right)^{\circ}\text{C}=41.3^{\circ}\text{C}$.

73. Required average = $\frac{(15\times 80)-[(16\times 15)+(14\times 25)]}{80-(15+25)}$

$=\frac{1200-(240+350)}{40}=\frac{610}{40}=15.25$.

74. Required average = ₹ $\left[\frac{(5680\times 75)-\{(5400\times 25)+(5700\times 30)\}}{75-(25+30)}\right]$

= ₹ $\left[\frac{426000-(135000+171000)}{20}\right]$

= ₹ $\left(\frac{120000}{20}\right)$ = ₹ 6000.

75. Required average = $\frac{(76\times 16)-(75\times 10)}{6}=\left(\frac{1216-750}{6}\right)$

$=\frac{466}{6}=\frac{233}{3}=77\frac{2}{3}$.

76. Let the highest marks obtained by the student be x.

Then, second highest marks = x – 2.

Sum of marks of these 2 subjects = (87 × 8) – (85 × 6) = 696 – 510 = 186.

$\therefore$ $x + (x - 2) = 186 \Rightarrow 2x = 188 \Rightarrow x = 94$.

77. Let the highest score be x. Then, lowest score = $(x - 172)$.

Then, (50 × 40) – [x + (x – 172)] = 38 × 48

$\Leftrightarrow$ $2x$ = 2000 + 172 – 1824

$\Leftrightarrow$ $2x$ = 348

$\Leftrightarrow$ x = 174.

78. Sum of ages of the captain and the youngest player

= [(28 × 11) – {(25 × 3) + (28 × 3) + (30 × 3)] years

= (308 – 249) years = 59 years.

Let the age of the youngest player be x years. Then, age of the captain = (x + 11) years.

$\therefore$ $x + (x + 11) = 59 \Rightarrow 2x = 48 \Rightarrow x = 24$.

79. Total price of the two books = ₹ [(12 × 10) – (11.75 × 8)] = ₹ (120 – 94) = ₹ 26.

Let the price of one book be ₹ x.

Then, the price of other book = ₹ (x + 60% of x)

= ₹ $\left(x+\frac{3}{5}x\right)$ = ₹ $\left(\frac{8x}{5}\right)$.

So, $x+\frac{8x}{5}=26 \Leftrightarrow 13x = 130 \Leftrightarrow x = 10$.

$\therefore$ The prices of the two books are ₹ 10 and ₹ 16.

80. Required number = (12 × 6) – (10 × 5) = 72 – 50 = 22.

81. Average after 11 innings = 36.

$\therefore$ Required number of runs = (36 × 11) – (32 × 10) = 396 – 320 = 76.

82. Total sale for 5 months = ₹ (6435 + 6927 + 6855 + 7230 + 6562) = ₹ 34009.

$\therefore$ Required sale = ₹ [(6500 × 6) – 34009]

= ₹ (39000 – 34009) = ₹ 4991.

83. Required average = $\frac{(4375\times 12)-(4000\times 3)}{9}$

$=\frac{52500-12000}{9}=\frac{40500}{9}=4500$.

84. Required run rate = $\frac{282-(3.2\times 10)}{40}=\frac{250}{40}=6.25$.

85. Let the average score of the remaining 6 batsmen be x runs.

Then, sum of their scores = $6x$; captain's score = (x + 30).

$\therefore$ $6x + (x + 30) = 310 \Rightarrow 7x = 280 \Rightarrow x = 40$.

Hence, captain's score = x + 30 = 70.

86. Number of men = $\left(\frac{1}{5}\times 120\right)=24$.

Number of women = $\left(\frac{1}{4}\times 120\right)=30$.

Number of children = 120 – (24 + 30) = 66.

Average age of men = 60 years.

Average age of children = $\left(\frac{1}{4}\times 60\right)$ years = 15 years.

Average age of women = $\left(\frac{5}{6}\times 60\right)$ years = 50 years.

$\therefore$ Average age of the group

$=\left(\frac{60\times 24+50\times 30+15\times 66}{120}\right)$ years

$=\left(\frac{3930}{120}\right)$ years = 32.75 years.

87. Let the attendance on the three days be $2x$, $5x$ and $13x$ respectively.

Then, total charges

= ₹ (15 × $2x$ + 7.50 × $5x$ + 2.50 × $13x$)

= ₹ ($30x$ + $37.5x$ + $32.5x$)

= ₹ 100 x.

$\therefore$ Average charge per person = ₹ $\left(\frac{100\,x}{2x+5x+13x}\right)$ = ₹ 5.